

# Machine, Data and Learning

# Linear Programming

- Mathematical programming is used to find the **best or optimal solution** to a problem that requires a decision or set of decisions about how best to use a set of limited resources to achieve a state goal of objectives.
- **Steps involved in mathematical programming**
  - Conversion of stated problem into a mathematical model that abstracts all the essential elements of the problem.
  - Exploration of different solutions of the problem.
  - Finding out the most suitable or optimum solution.
- **Linear programming** requires that all the mathematical functions in the model be **linear functions**.

# The Linear Programming Model (1)

Let:  $x_1, x_2, x_3, \dots, x_n$  = decision variables

$Z$  = Objective function or linear function

Requirement: Maximization of the linear function  $Z$ .

$$Z = c_1x_1 + c_2x_2 + c_3x_3 + \dots + c_nx_n \quad \dots \text{Eq (1)}$$

subject to the following constraints:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2$$

$$\vdots$$
$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_n$$

$$\text{all } x_j \geq 0$$

## The Linear Programming Model (2)

- The linear programming model can be written in more efficient notation as:

Maximize

$$Z = \sum_{j=1}^n c_j x_j$$

subject to:

$$\sum_{j=1}^n a_{ij} x_j \leq b_i$$

where

$$i = 1, 2, \dots, m$$

and

$$x_j \geq 0$$

where

$$j = 1, 2, \dots, n$$

The decision variables,  $x_1, x_2, \dots, x_n$ , represent levels of  $n$  competing activities.

# Examples of LP Problems (1)

## 1. A Product Mix Problem

- A manufacturer has fixed amounts of different resources such as raw material, labor, and equipment.
- These resources can be combined to produce any one of several different products.
- The quantity of the  $i^{th}$  resource required to produce one unit of the  $j^{th}$  product is known.
- The decision maker wishes to produce the combination of products that will **maximize total income**.

# Examples of LP Problems (2)

## 2. A Transportation Problem

- A product is to be shipped in the amounts  $a_1, a_2, \dots, a_m$  from  $m$  shipping origins and received in amounts  $b_1, b_2, \dots, b_n$  at each of  $n$  shipping destinations.
- The cost of shipping a unit from the  $i^{\text{th}}$  origin to the  $j^{\text{th}}$  destination is known for all combinations of origins and destinations.
- The problem is to determine the amount to be shipped from each origin to each destination such that the total **cost of transportation is a minimum.**

# Examples of LP Problems (3)

## 3. A Flow Capacity Problem

- One or more commodities (e.g., traffic, water, information, cash, etc.) are flowing from one point to another through a network whose branches have various constraints and flow capacities.
- The direction of flow in each branch and the capacity of each branch are known.
- The problem is to determine the **maximum flow**, or capacity of the network.

# Developing LP Model

- Consider the product mix problem
- **Steps Involved:**
  - Determine the objective of the problem and describe it by a criterion function in terms of the decision variables.
  - Find out the constraints.
  - Do the analysis which should lead to the selection of values for the decision variables that optimize the criterion function while satisfying all the constraints imposed on the problem.

# Developing LP Model

XYZ Company produces two products: I and II. The raw material requirements, space needed for storage, production rates, and selling prices for these products are given in Table 1.

**TABLE 1 Production Data for N. Dustrious Company**

|                                       | Product |    |
|---------------------------------------|---------|----|
|                                       | I       | II |
| Storage space (ft <sup>2</sup> /unit) | 4       | 5  |
| Raw material (lb/unit)                | 5       | 3  |
| Production rate (units/hr)            | 60      | 30 |
| Selling price (\$/unit)               | 13      | 11 |

The total amount of raw material available per day for both products is 15751b. The total storage space for all products is 1500 ft<sup>2</sup>, and a maximum of 7 hours per day can be used for production.

# Developing LP Model

All products manufactured are shipped out of the storage area at the end of the day. Therefore, the two products must share the total raw material, storage space, and production time. **The company wants to determine how many units of each product to produce per day to maximize its total income.**

## Solution

- The company has decided that it wants to maximize its sale income, which depends on the number of units of product I and II that it produces.
- Therefore, the decision variables,  $x_1$  and  $x_2$  can be the number of units of products I and II, respectively, produced per day.

# Developing LP Model

- The object is to maximize the equation:

$$Z = 13x_1 + 11x_2$$

subject to the constraints on storage space, raw materials, and production time.

- Each unit of product I requires 4 ft<sup>2</sup> of storage space and each unit of product II requires 5 ft<sup>2</sup>. Thus a total of  $4x_1 + 5x_2$  ft<sup>2</sup> of storage space is needed each day. This space must be less than or equal to the available storage space, which is 1500 ft<sup>2</sup>.

Therefore,

$$4X_1 + 5X_2 \leq 1500$$

- Similarly, each unit of product I and II produced requires 5 and 3 lbs, respectively, of raw material. Hence a total of  $5x_1 + 3x_2$  lb of raw material is used.

## Developing LP Model

- This must be less than or equal to the total amount of raw material available, which is 1575 lb. Therefore,

$$5x_1 + 3x_2 \leq 1575$$

- Product I can be produced at the rate of 60 units per hour. Therefore, it must take  $1/60$  of an hour to produce 1 unit. Similarly, it requires  $1/30$  of an hour to produce 1 unit of product II. Hence a total of  $x_1/60 + x_2/30$  hours is required for the daily production. This quantity must be less than or equal to the total production time available each day. Therefore,

$$x_1 / 60 + x_2 / 30 \leq 7 \text{ or } x_1 + 2x_2 \leq 420$$

- Finally, the company cannot produce a negative quantity of any product, therefore  $x_1$  and  $x_2$  must each be greater than or equal to zero.

# Developing LP Model

- The linear programming model for this example can be summarized as:

Maximize

$$Z = 13x_1 + 11x_2$$

subject to:

$$4x_1 + 5x_2 \leq 1500$$

$$5x_1 + 3x_2 \leq 1575$$

$$x_1 + 2x_2 \leq 420$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

# Linear Programming for MDPs

- Maximize

$$\sum_{i \in S} v_i \quad (S \text{ is the set of states, } v_i \text{ is value of state})$$

Such that:

$$v_i \leq [R(I,A) + \gamma \sum P(J|I,A) * v_j]$$

# Linear Programming

- Linear programming polynomial time (Karmarkar, 84)
  - Popular method Dantzig's simplex (1963)
  - Simplex performs well in practice, but could be exponential
- Although polynomial time typically slower than MDP specific algorithms such as value iteration
- Determine policy from  $V_i$  using method from previous slides
  - Yields a deterministic policy for MDPs: why?
- Key advantage: Many times it is amenable to modeling specific constraints

# A more popular formulation

$$\max \sum_i \sum_a x_{ia} r_{ia}$$

$$\sum_a x_{ja} - \sum_i \sum_a x_{ia} p_{ij}^a = \alpha_j,$$

$$x_{ia} \geq 0$$

$x_{ia}$ : Expected number of times action  $a$  is taken in state  $i$

$r_{ia}$ : Reward for taking action  $a$  in state  $i$

$p^a_{ij}$ : Probability of reaching state  $j$  when action  $a$  is taken in state  $i$

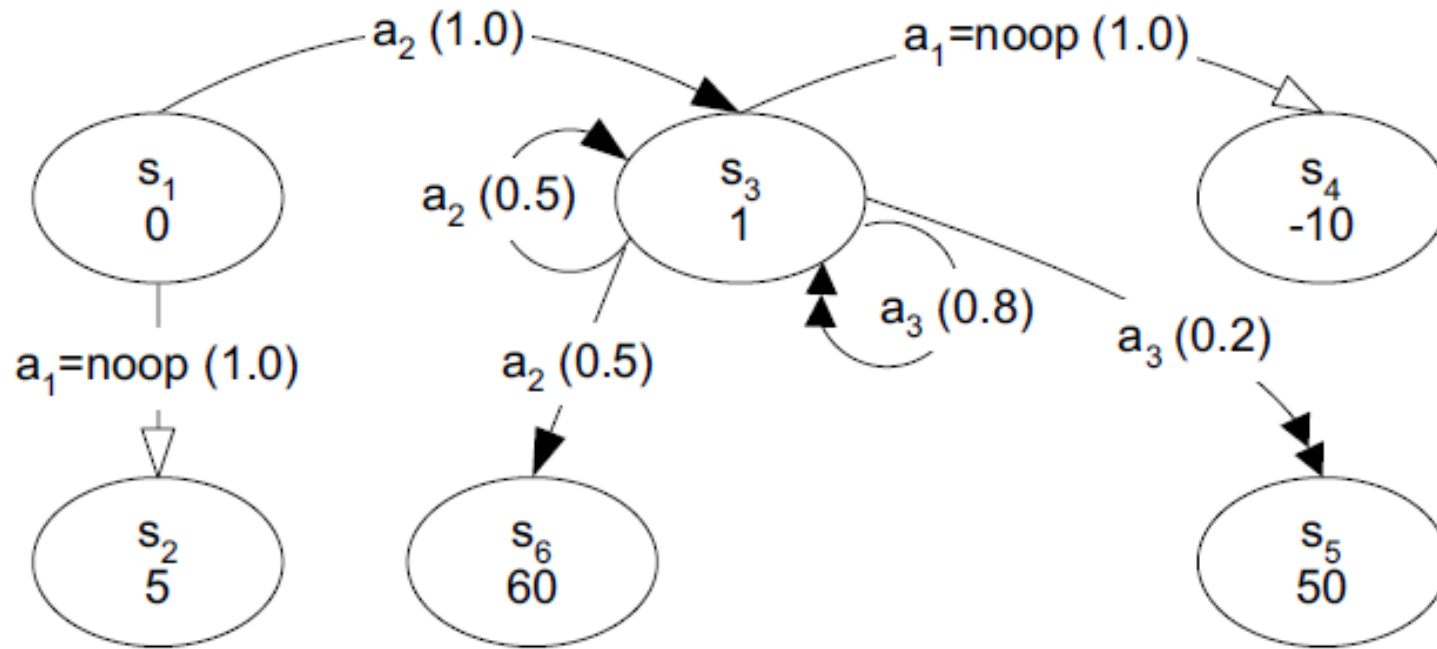
$\alpha_j$ : Initial probability of being in state  $j$

# Refactoring

$$\max \sum_i \sum_a x_{ia} r_{ia} \quad \left| \quad \begin{aligned} \sum_i \sum_a (\delta_{ij} - p_{ij}^a) x_{ia} &= \alpha_j, \\ x_{ia} &\geq 0, \end{aligned} \right.$$

where  $\delta_{ij}$  is the Kronecker delta, defined as  $\delta_{ij} = 1 \iff i = j$ .

# MDP Example



A simple MDP:  $S = \{s_1, \dots, s_6\}$ ,  $A = \{a_1 = \text{noop}, a_2, a_3\}$

# MDP Example

$$\max(\mathbf{r}\mathbf{x}) \mid \mathbf{A}\mathbf{x} = \boldsymbol{\alpha}, \mathbf{x} \geq 0,$$

$$\mathbf{x} = [(x_{11}, x_{12}), x_{21}, (x_{31}, x_{32}, x_{33}), x_{41}, x_{51}, x_{61}]^T,$$

$$\mathbf{r} = [(0, 0), 5, (1, 1, 1), -10, 50, 60], \quad \boldsymbol{\alpha} = [0.1, 0.1, 0.1, 0.1, 0.1, 0.5]^T,$$

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0.5 & 0.2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -0.2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -0.5 & 0 & 0 & 0 & 1 \end{pmatrix}$$

# Solution to the LP

$$\mathbf{x} = [(0, 0.1), 0.1, (0, 0.4, 0), 0.1, 0.1, 0.7]$$

$$\mathbf{d} = [a_2, a_1, a_2, a_1, a_1, a_1]$$

Solving the LP we get  $\mathbf{x}$ , which maps to a deterministic uniformly-optimal policy

We get the following output if we change alpha to the following

$$\alpha = [1, 0, 0, 0, 0, 0]$$

$$\mathbf{x} = [(0, 1), 0, (0, 2, 0), 0, 0, 1]$$

$$d_1 = a_2, d_3 = a_2, d_6 = a_1$$

arbitrary actions for  $s_2$ ,  $s_4$ , and  $s_5$

# CMDP

- A key advantage of the formulation is the ability to model constraints
- If we want to say state 1 must have all actions take with equal probability
  - $x_{11} = x_{12}$
- Addition of such constraints results in a CMDP (Constrained MDP)
- How do you model action 1 in state 1 must be taken 30% of time ?
  - $X_{11}/(x_{11}+x_{12}) = 1/3$
  - How can you do this with value or policy iteration ?

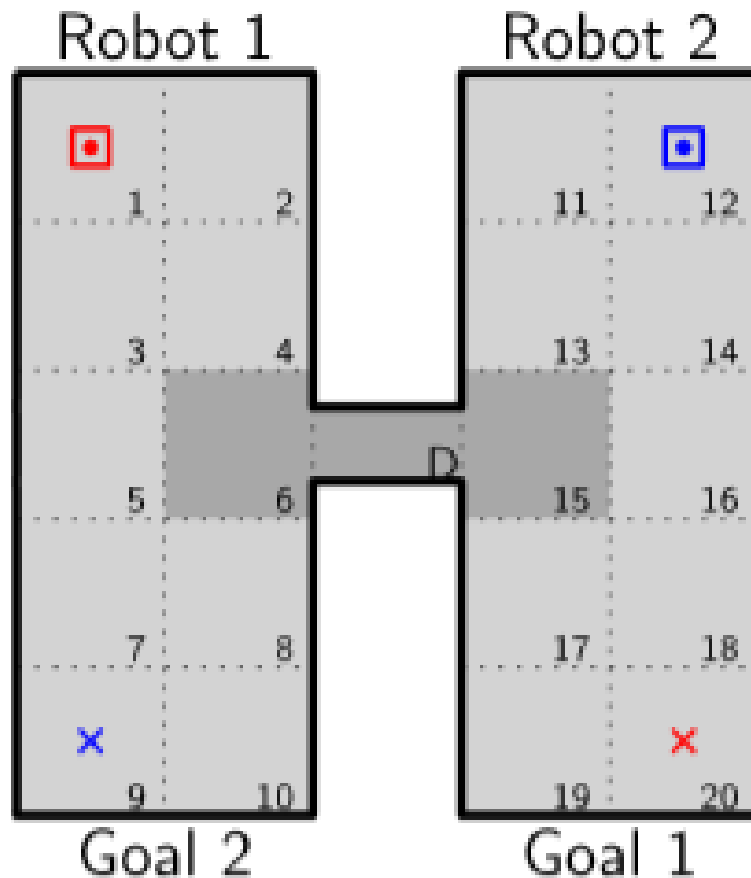
# Multi-Agent MDP (MMDP)

- **MMDP:** A direct extension over MDPs for multiple agents
- $M = \langle S, \{A_i\}_{i \in m}, T, R \rangle$ 
  - $S$  is set of possible world states
  - $\{A_i\}_{i \in m}$  is set of joint actions,  $\langle a_1, \dots, a_m \rangle$  where  $a_i \in A_i$
  - $T$  defines transition probabilities over joint actions
  - $R$  is team reward function
- State is fully observable by each agent
- In absence of communication, random policies can lead to mis-coordination
- Ex: In state  $s$ , policy  $a_1b_2 = .7$  and  $a_2b_1 = .3$  would lead to  $a_1b_1$  with .21,  $a_1b_2$  with .49,  $a_2b_1$  with .09 and  $a_2b_2$  with .21

# Dec-MDP

- In MMDP model agent observes the joint state
- Many times in a joint problem, an agent may observe only his/her local state
- Separating out the policy computation for each agent is not an option since they have joint transitions and rewards
- Dec-MDP – A formal framework to address these issues

# Robots coordinating in a hallway with a narrow passage



Can be modeled  
using Dec-MDP  
framework